

My white board writing from 8/08/2017

This document is not going to be very lengthy because the notes are very detailed.

I first discussed the proof of the important property 4 of MVN. All steps are outlined clearly in the notes. Same holds also about Property 5.

I stressed on the interpretation of Property 4. Assume that an output random variable Y depends on an input random vector $X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}$ and that $\begin{pmatrix} Y \\ X \end{pmatrix} \sim N_{p+1}(\mu, \Sigma)$ $(p+1)$ -dim normal random vector. Denote by C the covariance matrix of X , and by σ_0 the vector $\sigma_0 = \begin{pmatrix} \text{cov}(Y, X_1) \\ \text{cov}(Y, X_2) \\ \vdots \\ \text{cov}(Y, X_p) \end{pmatrix}$. Then obviously $\mu = \begin{pmatrix} EY \\ E(X) \end{pmatrix}$ and

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} = \begin{pmatrix} \text{Var } Y & \sigma_0' \\ \sigma_0 & C \end{pmatrix} \text{ holds.}$$

As we know that the best prediction of Y using functional transformation of X is attained by $E(Y/X)$ (see my white board writing from week 1) and since Property 4 gives this conditional expected value as $EY + \Sigma_{12} \Sigma_{22}^{-1} (X - E(X)) = EY + \sigma_0' C^{-1} (X - E(X)) =$
 $= b + \sigma_0' C^{-1} X$, with $b = EY - \sigma_0' C^{-1} E(X)$

This is a linear transformation of the components of the input. Denoting $C^{-1} \sigma_0 = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix}$ and $b = \beta_0$, it has the familiar form $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$

The function $E(Y|X)$ is called the regression function. We have just shown that for the case of joint multivariate normality of $\begin{pmatrix} Y \\ X \end{pmatrix}$ the regression function happens to be linear in the output, i.e. the best approximation of Y in quadratic error minimization sense, i.e. $\min_{g: E g(X)^2 < \infty} E(Y - g(X))^2$,

which is attained at $g^*(X) = E(Y|X)$, has a simple linear form $g^*(X) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$ Under this multivariate normality assumption.

In addition, the minimal mean-squared prediction error (again due to Property 4) is

$$\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} = \left| \text{Var } Y - \sigma_0' C^{-1} \sigma_0 \right|$$

Note that if Y and the vector X happen to be uncorrelated (i.e. $\text{cov}(Y, X_i) = 0, i = 1, 2, \dots, p$) then σ_0 will be a zero vector, hence the error above reduces to just $\text{Var } Y$. This would be the same value which we get when predicting Y by a constant EY (i.e., when disregarding the input).
Indeed $E[Y - EY]^2 = \text{Var } Y$.

The interpretation: in this case X and Y are also independent (Since jointly multivariate normal and uncorrelated) $\rightarrow$ so knowing a value of X is not expected to improve our prediction of Y and the variability of Y will not be reduced (stays = $\text{Var}(Y)$).